Some Simple Explicit Examples of Correlated One Electron Theory and Density Matrix Functionals

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Abstract

Some examples are presented displaying how General Spin Orbitals (GSO's) and their occupation numbers uniquely determine N-electron Antisymmetrized Geminal Power states. This provides a correlated one electron model that generalizes the independent particle model that is universally used in picturing chemical behavior. These examples also demonstrate that one electron models, involving GSO's and occupation numbers, and ones based on a first order density matrix are not equivalent. This difference is shown by displaying sets of different GSO's that have the same occupation numbers that correspond to the same density matrix but different N-electron states. This observation also leads to the explicit construction of a density matrix functional.

I. INTRODUCTION

The utility and power of pictorial models in the understanding and manipulation of complex physical phenomena is clearly evident in all areas of scientific endeavour. Perhaps one of the most ubiquitous examples of this is the use of molecular orbitals in understanding structure, properties and reactivity in chemistry and material science. This efficacy is due to the approximate validity of the Independent Particle State (IPS) model, where

- the properties of individual electrons and their effective interactions *determine uniquely* the many electron state and its properties
- electrons can be identified with orbitals and their observable quantities (e.g. energy, ionization potential, electron affinity etc..) in a one to one fashion
- orbitals provide a geometric representation of electrons and thus multi-electron states.

The IPS model, however, can predict incorrect properties, reactions and structures due to the absence of electron-electron correlation effects. In order to overcome this defect more accurate methods such as Configuration Interaction, Coupled Cluster, Perturbation Theory and many others have been utilized that contain an explicit description of these interactions. Although the resulting improved multi-electron states still determine sets of orbitals, now greater than the number of electrons and weighted with occupation numbers, through the First Order Reduced Density Operator (FORDO), these orbitals and their occupation numbers *do not*, in general, determine a unique multi-electron state. Thus one loses the concrete pictorial aspects of a one-electron theory, as the multi-electron state *cannot* be described solely in terms of single electron quantities and geometric forms.

Maintaining a one-electron picture while including electron correlation effects, i.e. constructing a correlated one-electron picture, is a central driving force in Kohn-Sham Density Functional Theory (KSDFT), Density Matrix Functional Theory (DMFT) and various quasi particle approaches. A common drawback in all of these methods is that it is not known how to express the multi-electron state as an explicit functional of orbitals and occupation numbers. Originally the orbitals produced in KS-DFT were introduced as mathematical artifacts, while recently [1] there has been some discussion about their physical meaning as correlated orbitals but still their connection to the N-electron state is not clear. However, there is a class of correlated multi-electron states and hence their energies that can be expressed exactly as known functionals of orbitals and occupation numbers, they are the Antisymmetrized Geminal Power (AGP) States [2].

The analysis of AGP states also clearly shows that a model based on orbitals and occupation numbers is not equivalent to one based on FORDOs, as many sets of orbitals determine the same FORDO while determining different N-electron states. This observation allows one to explicitly construct examples of DMFs whose domain is restricted to AGPs.

In this paper we review the definition of a correlated one-particle theory (section II), the structure and parameterization of AGP states, their N-electron energy functional, the expression of their energy in terms of one-electron quantities and discuss how one can interpret them pictorially (section III). In section IV we define the AGP DMF and in section V we describe simple examples of (i) AGP states that are determined by orbitals and occupation numbers and (ii) AGP DMFs and we conclude with a discussion in section VI.

As usual with AGPs, the presentation is restricted to systems with even number of electrons. Extensions to odd numbers is sketched in section VI. A further caveat a reader should keep in mind is although an exact quantum mechanical description of physical systems would require infinite dimensional Hilbert spaces one generally considers finite dimensional approximations. Thus the following material is expressed in terms of and refers to finite dimensional Hilbert subspaces, with the understanding that appropriate limits must be taken to obtain the exact infinite dimensional description.

II. ONE-ELECTRON THEORY

In order to be unambiguous we introduce the following definition: a one-electron Theory (OET) is one in which the

- N-electron ground state is *totally determined* by a set of orthonormal Generalized Spin Orbitals (GSOs), (see Appendix A), $\{|\psi_j\rangle\,; 1 \leq j \leq r; r \geq N\}$ spanning one-particle space and a set of occupation numbers $\{N_j; 1 \leq j \leq r; r \geq N\}$ of these GSOs

  and

- N-electron state energy can be expressed in terms of one-electron observables e.g. GSO kinetic energies, ionization potentials etc...

A OET is *distinct* from ones based on FORDO's, as different sets of GSO's (with the same occupation numbers) can produce the *same* FORDO while forming different N-electron states, with different energies. This difference can be understood by considering the action of the subgroup $U(\boldsymbol{\theta})$ of the unitary group $U(\mathcal{H}^1)$ of unitary transformations of one-electron space, $\mathcal{H}^1$, defined by

$$\begin{aligned} u(\boldsymbol{\theta})(|\psi_1\rangle, \cdots, |\psi_r\rangle) &= (|\psi_1\rangle, \cdots, |\psi_r\rangle) \begin{pmatrix} e^{i\theta_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & e^{i\theta_r} \end{pmatrix} \\ &= \left(e^{i\theta_1}|\psi_1\rangle, \cdots, e^{i\theta_r}|\psi_r\rangle\right) \end{aligned} \tag{1}$$

first on the FORDO D^1

$$D^1 = \sum_{1\le j\le r} N_j |\psi_j\rangle\langle\psi_j|, \tag{2}$$

then on the N-electron state density operator D^N (assuming a pure state)

$$D^N = \sum_{\substack{1\le j_1<\cdots<j_N\le r \\ 1\le k_1<\cdots<k_N\le r}} \bar{c}_{j_1\cdots j_N} c_{k_1\cdots k_N} |\psi_{j_1}\cdots\psi_{j_N}\rangle\langle\psi_{k_1}\cdots\psi_{k_N}|, \tag{3}$$

that contracts to D^1, where $\{c_{j_1\cdots j_N} | 1\le j_1<\cdots<j_N\le r\}$ are the CI coefficients of the state vector forming D^N. One should note that D^N is not in general diagonal in the basis $\{|\psi_{j_1}\cdots\psi_{j_N}\rangle | 1\le j_1<\cdots<j_N\le r\}$ produced by the natural GSOs $\{\psi_j | 1\le j\le r\}$ of D^1. The subgroup $U(\boldsymbol{\theta})$ leaves D^1 invariant as

$$u(\boldsymbol{\theta}) D^1 u(\boldsymbol{\theta})^\dagger = \sum_{1\le j\le r} N_j e^{i\theta_j}|\psi_j\rangle\langle\psi_j| e^{-i\theta_j} = \sum_{1\le j\le r} N_j|\psi_j\rangle\langle\psi_j|, \tag{4}$$

but it does not leave the state D^N invariant (the off diagonal terms are changed) as

$$\begin{aligned} u(\boldsymbol{\theta}) D^N u(\boldsymbol{\theta})^\dagger = &\sum_{\substack{1\le j_1<\cdots<j_N\le r \\ 1\le k_1<\cdots<k_N\le r}} \bar{c}_{j_1\cdots j_N} c_{k_1\cdots k_N} e^{i\left\{\theta_{j_1}+\cdots+\theta_{j_N}\right\}} e^{-i\left\{\theta_{k_1}+\cdots+\theta_{k_N}\right\}} |\psi_{j_1}\cdots\psi_{j_N}\rangle\langle\psi_{k_1}\cdots\psi_{k_N}| \\ &\neq D^N. \end{aligned} \tag{5}$$

The non invariance of the N-electron state density operator will also be shown in the following by simple examples that explicitly display the lack of invariance of the Second Order Reduced Density Operator's (SORDO's) of AGP states (thus any higher order density operators) to such transformations.

The observation that many sets of GSO's map into the same FORDO while producing different N-electron states allows one to construct concrete examples of DMF's (which are functionals of matrix representations of FORDO's) through a Levy-Lieb [3, 4] type of construction. Simple examples of such functionals will also be displayed in the following discussion.

III. ANTISYMMETRIZED GEMINAL POWER STATES

The most general form of an AGP state is produced by forming an antisymmetrized power of a single geminal composed of GSO's. Geminals can be expanded uniquely in parametric form as

$$|g(\boldsymbol{c}, \boldsymbol{\psi}, \boldsymbol{\theta})\rangle = \sum_{1 \le j \le s} c_j e^{i\theta_j} |\psi_j \psi_{j+s}\rangle , \tag{6}$$

where $\{|\psi_j\rangle ; 1 \le j \le r = 2s\}$ is a complete orthonormal basis of GSO's, ψ_{j+s} is the GSO *paired* with the GSO ψ_j, the canonical coefficients $\mathbf{c}$ lie on the positive portion of the hypersphere

$$\sum_{1 \le j \le s} c_j^2 = 1; \; c_j \ge 0; 1 \le j \le s, \tag{7}$$

the angles $\boldsymbol{\theta} = (\theta_1, \cdots, \theta_s)$ satisfy

$$0 \le \theta_2, \cdots, \theta_s < 2\pi; \; \theta_1 = 0 \tag{8}$$

and the GSO's $\{\psi_j, \psi_{j+s}\}$ belong to a set of non equivalent pairs, $\mathcal{P}$, (see Appendix B and the next subsection). This pairing *can* have physical significance as in the BCS theory of superconductivity [5–7] where ψ_j has the opposite spin and angular momenta to ψ_{j+s}, but can also just reflect the basic antisymmetric properties of $|g(\boldsymbol{c}, \boldsymbol{\psi}, \boldsymbol{\theta})\rangle$.

A. Density Operators

The FORDO of an AGP is always at least doubly degenerate i.e.

$$D^1(\boldsymbol{c}, \boldsymbol{\psi}) = \sum_{1 \le j \le s} N_j(\boldsymbol{c}) \{|\psi_j\rangle\langle\psi_j| + |\psi_{j+s}\rangle\langle\psi_{j+s}|\} \tag{9}$$

and it is uniquely parameterized by $\{\boldsymbol{c}, \boldsymbol{\psi}\}$, where $\boldsymbol{\psi} \in \mathcal{P}$ are representatives of the equivalence classes, $[\boldsymbol{\psi}]$, of paired normalized GSOs. For non degenerate AGPs theses are defined by eq.(10)

$$\{\psi_a, \psi_b\} \sim \{\psi_c, \psi_d\} \Longleftrightarrow |\psi_a \psi_b\rangle \langle \psi_a \psi_b| = |\psi_c \psi_d\rangle \langle \psi_c \psi_d| \,. \tag{10}$$

For degenerate AGPs the equivalence is more involved and is discussed in Appendix B. The occupation numbers $\{N_j(\boldsymbol{c}); 1 \leq j \leq s\}$ are given in terms of $\left\{n_j = c_j^2; 1 \leq j \leq s\right\}$ in a one-one fashion [8] by

$$N_j(\boldsymbol{c}) = n_j S_{\frac{N}{2}-1}(\hat{\jmath}) \left(S_{\frac{N}{2}}\right)^{-1}, \tag{11}$$

where

$$S_{\frac{N}{2}} \equiv S_{\frac{N}{2}}(\boldsymbol{n}) = \sum_{1 \leq j_1 < \cdots < j_{\frac{N}{2}} \leq s} n_{j_1} \cdots n_{j_{\frac{N}{2}}}, \tag{12a}$$

$$S_{\frac{N}{2}-1}(\hat{\jmath}) = \sum_{\substack{1 \leq j_1 < \cdots < j_{\frac{N}{2}-1} \leq s \\ j_1, \cdots, j_{\frac{N}{2}-1} \neq j}} n_{j_1} \cdots n_{j_{\frac{N}{2}-1}}, \tag{12b}$$

$$S_0(\hat{\jmath}) = 1, \quad S_m(\hat{\jmath}) = 0, \quad m < 0, \quad 1 \leq j \leq s, \tag{12c}$$

and $\hat{\jmath}$ signifies that j is missing from the sums in eqs. $(12a)$ and $(12b)$. For example in the case $N = 2$

$$S_1 = \sum_{1 \leq j_1 \leq s} n_{j_1}; \quad S_0(\hat{\jmath}) = 1, \quad 1 \leq j \leq s. \tag{13}$$

The SORDO is given by

$$D^2(\boldsymbol{c}, \boldsymbol{\psi}, \boldsymbol{\theta}) = \left(S_{\frac{N}{2}}\right)^{-1} \left\{ \sum_{1 \leq j \leq s} n_j S_{\frac{N}{2}-1}(\hat{\jmath}) |\psi_j \psi_{j+s}\rangle \langle \psi_j \psi_{j+s}| + \right.$$
$$\left. \sum_{1 \leq j \neq k \leq s} c_j c_k e^{i(\theta_j - \theta_k)} S_{\frac{N}{2}-1}\left(\hat{\jmath}\hat{k}\right) |\psi_j \psi_{j+s}\rangle \langle \psi_k \psi_{k+s}| + \sum_{\substack{1 \leq j < k \leq r \\ k \neq j+s}} n_j n_k S_{\frac{N}{2}-2}\left(\hat{\jmath}\hat{k}\right) |\psi_j \psi_k\rangle \langle \psi_j \psi_k| \right\}, \tag{14}$$

where

$$S_{\frac{N}{2}-1}\left(\hat{\jmath}\hat{k}\right) = \sum_{\substack{1 \leq j_1 < \cdots < j_{\frac{N}{2}-1} \leq s \\ j_1, \cdots, j_{\frac{N}{2}-1} \neq j,k}} n_{j_1} \cdots n_{j_{\frac{N}{2}-1}}, \tag{15}$$

$$S_{\frac{N}{2}-2}\left(\hat{\jmath}\hat{k}\right) = \sum_{\substack{1 \leq j_1 < \cdots < j_{\frac{N}{2}-2} \leq s \\ j_1, \cdots, j_{\frac{N}{2}-1} \neq j,k}} n_{j_1} \cdots n_{j_{\frac{N}{2}-2}}, \tag{16}$$

$$S_0(\hat{\jmath}_1 \cdots \hat{\jmath}_p) = 1; \quad S_m(\hat{\jmath}_1 \cdots \hat{\jmath}_p) = 0, \quad m < 0; \quad 0 \leq p \leq s. \tag{17}$$

B. State Vector

For N even the N-electron state vector has the CI form

$$\left|g\left(\boldsymbol{c},\boldsymbol{\psi},\boldsymbol{\theta}\right)^{\frac{N}{2}}\right\rangle = \sum_{1\le j_1<\cdots<j_{\frac{N}{2}}\le s} c_{j_1}\cdots c_{j_{\frac{N}{2}}}\exp\left\{i\sum_{1\le\kappa\le\frac{N}{2}}\theta_{j_\kappa}\right\}\left|\psi_{j_1}\psi_{j_1+s}\cdots\psi_{j_{\frac{N}{2}}}\psi_{j_{\frac{N}{2}}+s}\right\rangle, \quad (18)$$

which explicitly displays the dependence of AGP states that have the same FORDO (those with common $(\boldsymbol{c},\boldsymbol{\psi})$) on the phases $\{\theta_{j_\kappa}|\, 1\le\kappa\le s\}$.

C. Energy Functional

The energy of an AGP state is given by

$$\mathcal{E}\left(\boldsymbol{c},\boldsymbol{\psi},\boldsymbol{\theta}\right)=\frac{\mathcal{H}\left(\boldsymbol{c},\boldsymbol{\psi},\boldsymbol{\theta}\right)}{S_{\frac{N}{2}}\left(\boldsymbol{n}\right)}=$$
$$\left\{\sum_{1\le j\le s} n_j S_{\frac{N}{2}-1}\left(\hat{j}\right)\left\{\langle\psi_j\left|h_1\psi_j\right.\rangle+\langle\psi_{j+s}\left|h_1\psi_{j+s}\right.\rangle+\langle\psi_j\psi_{j+s}|\,|\psi_j\psi_{j+s}\rangle\right\}\right.$$
$$+\sum_{1\le j\ne k\le s}\left\{c_jc_ke^{i(\theta_j-\theta_k)}S_{\frac{N}{2}-1}\left(\hat{j}\hat{k}\right)\langle\psi_j\psi_{j+s}|\,|\psi_k\psi_{k+s}\rangle\right.$$
$$\left.\left.+\frac{1}{2}n_jn_kS_{\frac{N}{2}-2}\left(\hat{j}\hat{k}\right)\langle\psi_j\psi_k|||\psi_j\psi_k\rangle\right\}\right\}\Big/\ S_{\frac{N}{2}}\left(\boldsymbol{n}\right). \quad (19)$$

The terms involving the coulomb interaction are defined as

$$\langle\psi_j\psi_k|||\psi_j\psi_k\rangle=\langle\psi_j\psi_k||\psi_j\psi_k\rangle+\langle\psi_j\psi_{k+s}||\psi_j\psi_{k+s}\rangle$$
$$+\langle\psi_{j+s}\psi_k||\psi_{j+s}\psi_k\rangle+\langle\psi_{j+s}\psi_{k+s}||\psi_{j+s}\psi_{k+s}\rangle \quad (20)$$

and one can express the one and two electron expectation values in terms of of space-spin wave functions with coordinates $(\boldsymbol{r},\boldsymbol{\xi})$ in Hartree atomic units as

$$\langle\psi_j\left|h_1\psi_j\right.\rangle=\int\psi_j^*\left(\boldsymbol{r},\boldsymbol{\xi}\right)\left\{-\frac{1}{2}\nabla_{\boldsymbol{r}}^2+V_{\mathrm{ext}}\left(\boldsymbol{r}\right)\right\}\psi_j\left(\boldsymbol{r},\boldsymbol{\xi}\right)d\boldsymbol{r}d\boldsymbol{\xi} \quad (21)$$

and

$$\langle\psi_j\psi_k||\psi_l\psi_m\rangle=\int\left\{\psi_j^*\left(\boldsymbol{r}_1,\boldsymbol{\xi}_1\right)\psi_k^*\left(\boldsymbol{r}_2,\boldsymbol{\xi}_2\right)\frac{1}{\|\boldsymbol{r}_1-\boldsymbol{r}_2\|}\psi_l\left(\boldsymbol{r}_1,\boldsymbol{\xi}_1\right)\psi_m\left(\boldsymbol{r}_2,\boldsymbol{\xi}_2\right)\right.$$
$$\left.-\ \psi_j^*\left(\boldsymbol{r}_1,\boldsymbol{\xi}_1\right)\psi_k^*\left(\boldsymbol{r}_2,\boldsymbol{\xi}_2\right)\frac{1}{\|\boldsymbol{r}_1-\boldsymbol{r}_2\|}\psi_l\left(\boldsymbol{r}_2,\boldsymbol{\xi}_2\right)\psi_m\left(\boldsymbol{r}_1,\boldsymbol{\xi}_1\right)\right\}d\boldsymbol{r}_1d\boldsymbol{r}_2d\boldsymbol{\xi}_1d\boldsymbol{\xi}_2 \quad (22)$$

respectively.

D. One-Electron Picture

One can canonically absorb the phase factors in eq.(6) into the GSOs by setting

$$|g(\boldsymbol{c},\boldsymbol{\psi},\boldsymbol{\theta})\rangle = \sum_{1\le j\le s} c_j e^{i\theta_j}\,|\psi_j\psi_{j+s}\rangle = \sum_{1\le j\le s} c_j \left|\tilde{\psi}_j\tilde{\psi}_{j+s}\right\rangle, \tag{23}$$

where

$$\begin{aligned}\left|\tilde{\psi}_j\right\rangle &= e^{i\theta_j/2}\,|\psi_j\rangle \\ \left|\tilde{\psi}_{j+s}\right\rangle &= e^{i\theta_j/2}\,|\psi_{j+s}\rangle\end{aligned} \tag{24}$$

showing that the geminal is *uniquely defined* by the GSOs $\left\{\left|\tilde{\psi}_j\right\rangle\middle| 1\le j\le r\right\}$ and the occupation numbers $\{N_j| 1\le j\le r\}$, (through the inverse of the function defined by eq. (11)), thus satisfying the first of the two criteria for a OET displayed in section II.

E. One-Electron Quantities

In [2] it was shown that for AGPs states, $\left|g\left(\boldsymbol{c}_\#,\tilde{\boldsymbol{\psi}}_\#\right)^{\frac{N}{2}}\right\rangle$ that optimize the energy functional eq. (19) i.e.

$$\mathcal{E}\left(\boldsymbol{c}_\#,\tilde{\boldsymbol{\psi}}_\#\right) = \min_{\boldsymbol{c},\tilde{\boldsymbol{\psi}}} \mathcal{E}\left(\boldsymbol{c},\tilde{\boldsymbol{\psi}}\right) \tag{25}$$

one can express the total N-electron energy in terms of the occupation numbers $\{N_j(\boldsymbol{c}_\#)| 1\le j\le r\}$, Ionization Potentials (IPs), $\left\{I\left(\tilde{\psi}_{\#j}\right)\middle| 1\le j\le r\right\}$ and external potentials plus kinetic energies $\left\{h_1\left(\tilde{\psi}_{\#j}\right)\middle| 1\le j\le r\right\}$ associated with the optimized GSOs $\left\{\left|\tilde{\psi}_{\#j}\right\rangle\middle| 1\le j\le r\right\}$. The tilde signifies that the phase factor dependence has been absorbed into the GSOs as shown in subsection III D. The IP, $I\left(\tilde{\psi}_{\#j}\right)$, is the difference between the optimized N-electron state energy, $\mathcal{E}\left(\boldsymbol{c}_\#,\tilde{\boldsymbol{\psi}}_\#\right)$, and the N-1 electron state energy (not optimized), $\mathcal{E}_{N-1}\left(\boldsymbol{c}_\#,\tilde{\boldsymbol{\psi}}_\#,\tilde{\psi}_{\#j}\right)$, formed by removing the the j^{th} GSO from the set of GSOs that determines the N-electron energy, and is given by

$$I\left(\tilde{\psi}_{\#j}\right) = \mathcal{E}\left(\boldsymbol{c}_\#,\tilde{\boldsymbol{\psi}}_\#\right) - \mathcal{E}_{N-1}\left(\boldsymbol{c}_\#,\tilde{\boldsymbol{\psi}}_\#,\tilde{\psi}_{\#j}\right) \tag{26}$$

i.e. the frozen IP potential of the GSO $\left|\tilde{\psi}_{\#j}\right\rangle$. The energy $\mathcal{E}_{N-1}\left(\boldsymbol{c}_\#,\tilde{\boldsymbol{\psi}}_\#,\tilde{\psi}_{\#j}\right)$ is defined in terms of second quantized operators $\left\{a_j^\dagger\middle| 1\le j\le r=2s\right\}$ that create the GSO's

$\left\{ \tilde{\psi}_{\#j} \middle| 1 \leq j \leq r = 2s \right\}$ by

$$\mathcal{E}_{N-1}\left(\boldsymbol{c}_{\#}, \tilde{\boldsymbol{\psi}}_{\#}, \tilde{\psi}_{\#j}\right) = \frac{\left\langle g\left(\boldsymbol{c}_{\#}, \tilde{\boldsymbol{\psi}}_{\#}\right)^{\frac{N}{2}} \middle| a_j^{\dagger} H a_j g\left(\boldsymbol{c}_{\#}, \tilde{\boldsymbol{\psi}}_{\#}\right)^{\frac{N}{2}} \right\rangle}{\left\langle g\left(\boldsymbol{c}_{\#}, \tilde{\boldsymbol{\psi}}_{\#}\right)^{\frac{N}{2}} \middle| a_j^{\dagger} a_j g\left(\boldsymbol{c}_{\#}, \tilde{\boldsymbol{\psi}}_{\#}\right)^{\frac{N}{2}} \right\rangle}, \tag{27}$$

where H is the Hamiltonian defined by the operators whose matrix elements are given by eqs. (21) and (22). The one-electron expectation value $h_1\left(\tilde{\psi}_{\#j}\right)$ is the sum of the kinetic and external interaction energies and is given as

$$h_1\left(\tilde{\psi}_{\#j}\right) = \left\langle \tilde{\psi}_{\#j} \middle| h\tilde{\psi}_{\#j} \right\rangle = \left\langle \tilde{\psi}_{\#j} \middle| \left(K + V_{\text{ext}}\right) \tilde{\psi}_{\#j} \right\rangle. \tag{28}$$

The expression for the total energy is

$$\mathcal{E}\left(\boldsymbol{c}_{\#}, \tilde{\boldsymbol{\psi}}_{\#}\right) = \frac{1}{2} \sum_{1 \leq j \leq 2s} N_j\left(\boldsymbol{c}_{\#}\right) \kappa\left(\tilde{\psi}_{\#j}\right), \tag{29}$$

where

$$\kappa\left(\tilde{\psi}_{\#j}\right) = I\left(\tilde{\psi}_{\#j}\right) + h_1\left(\tilde{\psi}_{\#j}\right) \tag{30}$$

and $N_j\left(\boldsymbol{c}_{\#}\right)$ is the optimized occupation number of the GSO $\left|\tilde{\psi}_{\#j}\right\rangle$.

F. Orbital Models

The one-electron independent particle model is based on GSOs obtained by energy optimizing a single determinantal wave function, that have integer occupation numbers. As the number of electrons and the number of orthonormal GSOs $\{|\psi_j\rangle| 1 \leq j \leq N\}$ that define the IPS N-electron state are equal the magnitudes squared of the GSOs can be identified with the Probability Densities (PDs)

$$p_{\psi_j}\left(\boldsymbol{r}, \boldsymbol{\xi}\right) = \psi_j^*\left(\boldsymbol{r}, \boldsymbol{\xi}\right) \psi_j\left(\boldsymbol{r}, \boldsymbol{\xi}\right), \tag{31}$$

which can then interpreted as charge clouds. These GSOs are not unique as N-electron IPSs are invariant to the unitary group $U\left(N\right)$ that map the N-dimensional subspace generated by $\{|\psi_j\rangle| 1 \leq j \leq N\}$ to itself. Hence any set of orthonormal GSOs

$$\{u\,|\psi_j\rangle| 1 \leq j \leq N\}; \quad u \in U\left(N\right) \tag{32}$$

and their associated PDs, *which in general have different shapes*, give equally valid descriptions of the distribution of the N-electrons. In order to remove this pictorial ambiguity one can consider the optimized Fock operator, which is represented in the basis of optimized GSOs by a matrix which is divided into two blocks one $N \times N$ (the occupied GSOs) the other $(r-N) \times (r-N)$ (the virtual GSOs). Canonical occupied GSOs can then be defined as the eigenvectors of the occupied block by finding a $u \in U(N)$ that diagonalizes this block. The associated eigenvalues are ionization potentials associated with these canonical GSOs. As both the FORDO of the optimized IPS and the state itself are invariant to the group $U(N)$ the PDs produced by the canonical GSOs can then be used to form a standard pictorial description of the electronic distribution. However, from basic quantum mechanical considerations the canonical GSOs and the resulting PDs are not any more meaningful than any other set given by eq. (32). Other criteria, such as localization properties can and have been utilized that produce different visualizations, which are physically equivalent, but emphasize different aspects of the electronic structure. The importance of a specific canonical GSO to the reactivity of a given material/molecule can be estimated from the expression eq. (29), which can be expanded as

$$\mathcal{E}\left(\tilde{\boldsymbol{\psi}}_{\#}\right) = \frac{1}{2}\sum_{1\leq j\leq N} \kappa\left(\tilde{\psi}_{\#j}\right) = \frac{1}{2}\sum_{1\leq j\leq N}\left\{\epsilon\left(\tilde{\psi}_{\#j}\right) + h_1\left(\tilde{\psi}_{\#j}\right)\right\}, \tag{33}$$

where $\left\{\epsilon\left(\tilde{\psi}_{\#j}\right)\middle| 1 \leq j \leq N\right\}$ are the eigenvalues of the occupied GSOs of the Fock operator. The highest energy GSO $\left|\tilde{\psi}_{\#N}\right\rangle$, where $\kappa\left(\tilde{\psi}_{\#N}\right) \geq \cdots \geq \kappa\left(\tilde{\psi}_{\#1}\right)$, corresponds to the most reactive "electron", (note we have used the *non conventional* association between orbitals and energies given by $\left\{\kappa\left(\tilde{\psi}_{\#j}\right)\right\}$ which takes into account both the IP, effective coulombic interaction and the kinetic energy and can be deduced by the use of the virial theorem).

The OET introduced in section II generalizes this picture to correlated electrons described by AGP states, which contain IPSs as a special case. Now, however, the N-electron state is determined by more than N GSOs with non integer occupation numbers and thus one cannot associate an individual electron with a specific PD. It is still possible to visualize the structure in terms of electrons distributed according to the PDs produced by the representative canonical orbitals *of the optimized geminal*, but now weighted with the associated occupation numbers. The total energy can again be expressed as eq. (29) which can be recast in terms of the diagonal elements $\left\{\epsilon\left(\tilde{\psi}_{\#j}\right)\middle| 1 \leq j \leq r\right\}$ of a generalized Fock operator

[2] as

$$\mathcal{E}\left(\boldsymbol{c}_{\#},\tilde{\boldsymbol{\psi}}_{\#}\right)=\frac{1}{2}\sum_{1\leq j\leq s}N\left(\tilde{\psi}_{\#j}\right)\left\{\kappa\left(\tilde{\psi}_{\#j}\right)+\kappa\left(\tilde{\psi}_{\#j+s}\right)\right\}$$
$$=\frac{1}{2}\sum_{1\leq j\leq s}N\left(\tilde{\psi}_{\#j}\right)\left\{\epsilon\left(\tilde{\psi}_{\#j}\right)+\epsilon\left(\tilde{\psi}_{\#j+s}\right)+h_1\left(\tilde{\psi}_{\#j}\right)+h_1\left(\tilde{\psi}_{\#j+s}\right)\right\}. \tag{34}$$

The generalized Fock matrix is in general *non diagonal* i.e. $\epsilon\left(\tilde{\psi}_{\#j}\right)$ are not eigenvalues and is a *full* $r\times r$ matrix. The most reactive electron now corresponds to PDs $\{p_j|\, j\in\{j_{1H},\cdots,j_{mH}\}\}$, that are weighted by the occupation numbers $\left\{N\left(\tilde{\psi}_{\#j}\right)\middle|\, j\in\{j_{1H},\cdots,j_{mH}\}\right\}$, produced by the GSOs $\left\{\tilde{\psi}_{\#j}\middle|\, j\in\{j_{1H},\cdots,j_{mH}\}\right\}$, where

$$\sum_{j\in\{j_{1H},\cdots,j_{mH}\}}N\left(\tilde{\psi}_{\#j}\right)\kappa\left(\tilde{\psi}_{\#j}\right)=\max_{\substack{j\in\{j_1,\cdots,j_m\}\\ N\leq m\leq 2s}}\left\{\sum_{j_1,\cdots,j_m}N\left(\tilde{\psi}_{\#j}\right)\kappa\left(\tilde{\psi}_{\#j}\right)\right\}. \tag{35}$$

The set of subscripts $\{j_1,\cdots,j_m\}$ are determined by the constraint

$$\min_{\substack{\{j_1,\cdots,j_m\}\\ N\leq m\leq 2s}}\left|\sum_{j\in\{j_1,\cdots,j_m\}}N\left(\tilde{\psi}_{\#j}\right)-1\right|, \tag{36}$$

which cannot be expressed as

$$\sum_{j\in\{j_1,\cdots,j_m\}}N\left(\tilde{\psi}_{\#j}\right)-1=0 \tag{37}$$

as there may be no solution. The GSOs $\left|\tilde{\psi}_{\#j}\right\rangle$, $\left|\tilde{\psi}_{\#j+s}\right\rangle$, always have the same occupation numbers due to the pairing degeneracy, though the energies $\left\{\kappa\left(\tilde{\psi}_{\#j}\right),\kappa\left(\tilde{\psi}_{\#j+s}\right)\right\}$ might not be the same. The description of the next most energetic electron can be obtained in an analogous fashion.

IV. DENSITY MATRIX FUNCTIONAL

In this section we return to the non redundant parameterization of eq. (6) , which explicitly involves the phase. As the FORDO is left invariant by the change of phase, while the N-electron state and energy are not, one can define an AGP DMF (a functional defined over the domain of AGP states) by

$$\mathcal{F}\left(D^1\left(\boldsymbol{c},\boldsymbol{\psi}\right)\right)=\min_{\boldsymbol{\theta}}\mathcal{E}\left(\boldsymbol{\theta}\right) \tag{38}$$

where for fixed $\boldsymbol{c}$ and $\boldsymbol{\psi}$ the energy is just a function of the phases $\{\theta_j | \, 1 \leq j < k \leq s\}$ given by eq. (8), as

$$\mathcal{E}(\boldsymbol{\theta}) = \frac{\mathcal{H}(\theta)}{S_{\frac{N}{2}}} = S_{\frac{N}{2}}{}^{-1} \left\{ \sum_{1 \leq j \leq s} n_j S_{\frac{N}{2}-1}(\hat{\jmath}) \left\{ \langle \psi_j | h_1 \psi_j \rangle + \langle \psi_{j+s} | h_1 \psi_{j+s} \rangle + \langle \psi_j \psi_{j+s} | \, | \psi_j \psi_{j+s} \rangle \right\} \right.$$
$$\left. + \sum_{1 \leq j \neq k \leq s} \left\{ c_j c_k e^{i(\theta_j - \theta_k)} S_{\frac{N}{2}-1}\left(\hat{\jmath}\hat{k}\right) \langle \psi_j \psi_{j+s} | \, | \psi_k \psi_{k+s} \rangle + \frac{1}{2} n_j n_k S_{\frac{N}{2}-2}\left(\hat{\jmath}\hat{k}\right) \langle \psi_j \psi_k | | | \psi_j \psi_k \rangle \right\} \right\}. \tag{39}$$

V. SIMPLE EXAMPLES

In order to make the preceding discussion more tangible one can introduce very simple examples that involve only two electrons i.e. $N = 2$. These demonstrate

- how one-electron orbitals and occupation numbers determine correlated multi-electron (in this case two electron) states and
- the difference between an OET and a FORDO theory, which consequently leads to the construction of an explicit form for a DMF.

These subtle differences can even be observed when one restricts attention to spin orbitals or even orbitals for the two electron systems. Thus we can restrict attention to geminals that are low spin eigenfunctions of the spin operator $\mathcal{S}_0$ (see Appendix A 1) i.e.

$$\mathcal{S}_0 |g\rangle = 0 \tag{40}$$

formed by spin orbitals $\{|\varphi_1\alpha\rangle, \cdots, |\varphi_s\alpha\rangle, |\varphi_{s+1}\beta\rangle, \cdots, |\varphi_{2s}\beta\rangle\}$. The most general AGP of this type is in fact a geminal of the form

$$|g(\boldsymbol{c}, \boldsymbol{\varphi}, \boldsymbol{\theta})\rangle = \sum_{1 \leq j \leq s} c_j e^{i\theta_j} |\varphi_j \alpha \varphi_{j+s} \beta\rangle, \tag{41}$$

which describes correlated two electron states if $s \geq 2$.

A. Unrestricted Orbital Example

The simplest example of *different* states that have the *same* FORDO that are determined by spin orbitals and occupation numbers is given by eq. (41) when we set (i)

$$s = 2; \quad c_1 = \frac{1}{\sqrt{2}}, c_2 = \frac{1}{\sqrt{2}}; \quad \theta_1 = 0, \theta_2 = 0 \tag{42}$$

producing

$$|g_+\rangle = \frac{1}{\sqrt{2}} \{|\varphi_1 \alpha \varphi_3 \beta\rangle + |\varphi_2 \alpha \varphi_4 \beta\rangle\} \equiv \frac{1}{\sqrt{2}} \{|1\alpha 3\beta\rangle + |2\alpha 4\beta\rangle\} \tag{43}$$

and set (ii)

$$s = 2; \quad c_1 = \frac{1}{\sqrt{2}}, c_2 = \frac{1}{\sqrt{2}}; \quad \theta_1 = 0, \theta_2 = \pi \tag{44}$$

producing

$$|g_-\rangle = \frac{1}{\sqrt{2}} \{|1\alpha 3\beta\rangle - |2\alpha 4\beta\rangle\}, \tag{45}$$

where the succinct notation

$$j \equiv \varphi_j; \quad 1 \leq j \leq 2s \tag{46}$$

has been introduced. The common FORDO is

$$D^1 = \frac{1}{4} \{|1\alpha\rangle \langle 1\alpha| + |2\alpha\rangle \langle 2\alpha| + |3\beta\rangle \langle 3\beta| + |4\beta\rangle \langle 4\beta|\} \tag{47}$$

while the SORDOs are

$$D^2(g_+) = \frac{1}{2} \{|1\alpha 3\beta\rangle \langle 1\alpha 3\beta| + |1\alpha 3\beta\rangle \langle 2\alpha 4\beta| + |2\alpha 4\beta\rangle \langle 1\alpha 3\beta| + |2\alpha 4\beta\rangle \langle 2\alpha 4\beta|\} \tag{48}$$

and

$$D^2(g_-) = \frac{1}{2} \{|1\alpha 3\beta\rangle \langle 1\alpha 3\beta| - |1\alpha 3\beta\rangle \langle 2\alpha 4\beta| - |2\alpha 4\beta\rangle \langle 1\alpha 3\beta| + |2\alpha 4\beta\rangle \langle 2\alpha 4\beta|\}. \tag{49}$$

The relationship between the SORDOs eqs. (48, 49) and the general form of the SORDO is shown in Appendix C.

In the context of a OET (section II) the states $|g_+\rangle$ and $|g_-\rangle$ are determined by the representative spin orbitals and occupation numbers

$$\{\varphi_1\alpha, \varphi_2\alpha, \varphi_3\beta, \varphi_4\beta\}; \left\{\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right\} \tag{50}$$

and

$$\{\varphi_1\alpha, -\varphi_2\alpha, \varphi_3\beta, \varphi_4\beta\}; \left\{\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right\} \tag{51}$$

respectively. One should note that different representative sets of spin orbitals could be chosen from the equivalence classes $[\boldsymbol{\psi}]$ (recall the discussion at eq. (10)), but the phase dependence (in these cases the phases $\theta_1 = 0$ in eqs. (50, 51), $\theta_2 = 0$ in eq. (50) and $\theta_2 = \pi$ eq. (51)) *have to be included* in the description of the spin orbital.

These states have different energies which are given by

$$\mathcal{E}\left(g_{+}\right)=\frac{1}{2}\Big\{\left\langle 1\alpha\left|h_{1}1\alpha\right.\right\rangle+\left\langle 2\alpha\left|h_{1}2\alpha\right.\right\rangle+\left\langle 3\beta\left|h_{1}3\beta\right.\right\rangle+\left\langle 4\beta\left|h_{1}4\beta\right.\right\rangle$$
$$+\left\langle 1\alpha 3\beta\right|\left|1\alpha 3\beta\right\rangle+\left\langle 2\alpha 4\beta\right|\left|2\alpha 4\beta\right\rangle+\left\langle 1\alpha 3\beta\right|\left|2\alpha 4\beta\right\rangle+\left\langle 2\alpha 4\beta\right|\left|1\alpha 3\beta\right\rangle\Big\} \tag{52}$$

and

$$\mathcal{E}\left(g_{-}\right)=\frac{1}{2}\Big\{\left\langle 1\alpha\left|h_{1}1\alpha\right.\right\rangle+\left\langle 2\alpha\left|h_{1}2\alpha\right.\right\rangle+\left\langle 3\beta\left|h_{1}3\beta\right.\right\rangle+\left\langle 4\beta\left|h_{1}4\beta\right.\right\rangle$$
$$+\left\langle 1\alpha 3\beta\right|\left|1\alpha 3\beta\right\rangle+\left\langle 2\alpha 4\beta\right|\left|2\alpha 4\beta\right\rangle-\left\langle 1\alpha 3\beta\right|\left|2\alpha 4\beta\right\rangle-\left\langle 2\alpha 4\beta\right|\left|1\alpha 3\beta\right\rangle\Big\}\,. \tag{53}$$

B. Restricted Orbital Example

The preceding example can be made even simpler by considering restricted spin orbitals

$$\left|\varphi_{1}\right\rangle=\left|\varphi_{3}\right\rangle$$
$$\left|\varphi_{2}\right\rangle=\left|\varphi_{4}\right\rangle \tag{54}$$

leading to

$$\left|g_{\mathrm{res}+}\right\rangle=\left(\sqrt{2}\right)^{-1}\left\{\left|1\alpha 1\beta\right\rangle+\left|2\alpha 2\beta\right\rangle\right\}$$
$$\left|g_{\mathrm{res}-}\right\rangle=\left(\sqrt{2}\right)^{-1}\left\{\left|1\alpha 1\beta\right\rangle-\left|2\alpha 2\beta\right\rangle\right\}. \tag{55}$$

The FORDO, SORDOs and energies can be obtained from eqs. $(47-53)$ by replacing "3" with "1" and "4" with "2". The restricted geminals eq. (55) are now pure spin singlet states and can be factored into a product of a space part and spin part (see Appendix A 2).

Representative spin orbitals and occupation numbers determining the states $\left|g_{\mathrm{res}+}\right\rangle$ and $\left|g_{\mathrm{res}-}\right\rangle$ are

$$\left\{\varphi_{1}\alpha,\varphi_{2}\alpha,\varphi_{1}\beta,\varphi_{2}\beta\right\};\quad\left\{\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2}\right\}, \tag{56}$$
$$\left\{\varphi_{1}\alpha,i\varphi_{2}\alpha,\varphi_{1}\beta,i\varphi_{3}\beta\right\};\quad\left\{\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2}\right\} \tag{57}$$

respectively.

C. Density Matrix Functionals

Consider the manifold of geminals of the form

$$\left|g\left(\boldsymbol{c},\boldsymbol{\psi},\theta\right)\right\rangle=c_{1}\left|1\alpha 3\beta\right\rangle+c_{2}e^{i\theta}\left|2\alpha 4\beta\right\rangle, \tag{58}$$

where

$$
\begin{aligned}
&c_1^2 + c_2^2 = 1 \\
&0 \leq \theta < 2\pi.
\end{aligned} \tag{59}
$$

The constraint on the canonical coefficients leads to the parameterization

$$
\begin{aligned}
&c_1 = \cos\chi \\
&c_2 = \sin\chi \\
&0 \leq \chi \leq \frac{\pi}{2}
\end{aligned} \tag{60}
$$

and hence

$$
|g(\chi, \boldsymbol{\psi}, \theta)\rangle = \cos\chi\, |1\alpha 3\beta\rangle + e^{i\theta} \sin\chi\, |2\alpha 4\beta\rangle\,. \tag{61}
$$

All the states with a fixed $\{\chi, \boldsymbol{\psi}\}$, where $\boldsymbol{\psi} \in [\boldsymbol{\psi}]$, have a common FORDO given by

$$
D^1(\chi, \boldsymbol{\psi}) = \cos^2\chi \{|1\alpha\rangle\langle 1\alpha| + |3\beta\rangle\langle 3\beta|\} + \sin^2\chi \{|2\alpha\rangle\langle 2\alpha| + |4\beta\rangle\langle 4\beta|\}
$$

but different SORDOs

$$
\begin{aligned}
D^2(\chi, \boldsymbol{\psi}) = {} & \cos^2\chi\, |1\alpha 3\beta\rangle\langle 1\alpha 3\beta| + \sin^2\chi\, |2\alpha 4\beta\rangle\langle 2\alpha 4\beta| + \\
& \frac{1}{2}\left\{e^{-i\theta} |1\alpha 3\beta\rangle\langle 2\alpha 4\beta| + e^{i\theta} |2\alpha 4\beta\rangle\langle 1\alpha 3\beta|\right\} \sin 2\chi.
\end{aligned} \tag{62}
$$

These states have different energies that can be expressed by

$$
\begin{aligned}
\mathcal{E}(\theta) = {} & \{\langle 1\alpha\,|h_1 1\alpha\rangle + \langle 3\beta\,|h_1 3\beta\rangle\}\cos^2\chi + \{\langle 2\alpha\,|h_1 2\alpha\rangle + \langle 4\beta\,|h_1 4\beta\rangle\}\sin^2\chi \\
& + \langle 1\alpha 3\beta|\,|1\alpha 3\beta\rangle\cos^2\chi + \langle 2\alpha 4\beta|\,|2\alpha 4\beta\rangle\sin^2\chi \\
& + \frac{1}{2}\left\{e^{i\theta}\langle 1\alpha 3\beta|\,|2\alpha 4\beta\rangle + e^{-i\theta}\langle 2\alpha 4\beta|\,|1\alpha 3\beta\rangle\right\}\sin 2\chi.
\end{aligned} \tag{63}
$$

After spin integration eq. (63) gives

$$
\begin{aligned}
\mathcal{E}(\theta) = {} & \{(1\,|h_1 1) + (3\,|h_1 3)\}\cos^2\chi + \{(2\,|h_1 2) + (4\,|h_1 4)\}\sin^2\chi \\
& + (13|\,13)\cos^2\chi + (24|\,24)\sin^2\chi + \mathrm{Re}\left\{e^{i\theta}(13|\,24)\sin 2\chi\right\} \\
\equiv {} & k(\chi, \boldsymbol{\psi}) + f(\chi, \boldsymbol{\psi}, \theta)\,,
\end{aligned} \tag{64}
$$

where

$$
(jk|\,lm) = \int \varphi_j^*(\boldsymbol{r}_1)\,\varphi_k^*(\boldsymbol{r}_2)\,\frac{1}{\|\boldsymbol{r}_1 - \boldsymbol{r}_2\|}\varphi_l(\boldsymbol{r}_1)\,\varphi_m(\boldsymbol{r}_2)\,d\boldsymbol{r}_1 d\boldsymbol{r}_2, \tag{65}
$$

$$(j\,|h_1 j\,) = \int \varphi_j^*(\boldsymbol{r}) \left\{ -\frac{1}{2}\nabla_{\boldsymbol{r}}^2 + V_{\text{ext}}(\boldsymbol{r}) \right\} \varphi_j(\boldsymbol{r})\, d\boldsymbol{r}, \tag{66}$$

$$\begin{aligned} k(\chi,\boldsymbol{\psi}) = \{(1\,|h_1 1\,) + (3\,|h_1 3\,)\}\cos^2\chi + \{(2\,|h_1 2\,) + (4\,|h_1 4\,)\}\sin^2\chi \\ + (\,13|\,13)\cos^2\chi + (\,24|\,24)\sin^2\chi \end{aligned} \tag{67}$$

and

$$f(\chi,\boldsymbol{\psi},\theta) = \text{Re}\left\{e^{i\theta}(\,13|\,24)\sin 2\chi\right\}. \tag{68}$$

Over the domain of such geminals one can define a DMF as

$$\mathcal{F}(\chi,\boldsymbol{\psi}) = \mathcal{F}\left(D^1(\chi,\boldsymbol{\psi})\right) = \min_{\boldsymbol{\theta}} \mathcal{E}(\theta). \tag{69}$$

The only part of the functional $\mathcal{E}$ that depends on θ is $f(\chi,\boldsymbol{\psi},\theta)$, which one can express as

$$f(\chi,\boldsymbol{\psi},\theta) = \text{Re}\left\{e^{i\theta}(\,13|\,24)\right\} = \text{Re}\left\{e^{i\theta}ae^{i\alpha}\right\} = \text{Re}\left\{e^{i(\theta+\alpha)}a\right\} = a\cos(\theta+\alpha), \tag{70}$$

where

$$a = |(\,13|\,24)| \tag{71}$$

$$\alpha = \arg\{(\,13|\,24)\}. \tag{72}$$

The minimum of this function can be easily evaluated giving

$$f(\chi,\boldsymbol{\psi},\theta_{\text{min}}) = \min_{\theta}\{a\cos(\theta+\alpha)\} = -a \tag{73}$$

$$\theta_{\text{min}} = \pi - \alpha. \tag{74}$$

Hence the value of the DMF, $\mathcal{F}$, at the point $D^1(\chi,\boldsymbol{\psi})$ is

$$\mathcal{F}(\chi,\boldsymbol{\psi}) = k(\chi,\boldsymbol{\psi}) - 2\,|(\,13|\,24)|\sin 2\chi \tag{75}$$

and the minimum of the AGP energy functional $\mathcal{E}(\boldsymbol{c}(\chi),\boldsymbol{\psi},\theta)$, which determines the AGP ground state, is given by

$$\mathcal{E}(|g_{GS}\rangle) = \min_{\chi,\boldsymbol{\psi}} \mathcal{F}(\chi,\boldsymbol{\psi}). \tag{76}$$

VI. DISCUSSION

We have shown how the universally used orbital picture of electrons that was rigorously derived for IPSs i.e. for uncorrelated systems, but often used for correlated systems, can in fact be rigorously extended to a description of correlated ones. This allows one to maintain the single electron description of a multi-electron system and construct a pictorial model as outline in subsection III F. The manifold of N-electron states that are determined by GSOs and their occupation numbers are formed by AGP states, which describe systems that contain an even number of electrons. In order to describe systems with an odd number of electrons by a OET one can use Generalized AGP (GAGP) states [9] that are formed by taking an antisymmetrized product of a GSO with an AGP state. AGP states are not in general ground states for molecular systems, but can produce approximate ground states that have a lower energy than IPSs.

OETs are more general than FORDO theories as GSOs with the same occupation numbers can produce different N-electron states while producing the same FORDO. We constructed some simple two electron examples that displayed this feature, which we subsequently used to construct an example of an explicit DMF over the domain of geminals.

VII. ACKNOWLEDGEMENTS

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APPENDIX A: SPIN

1. General Spin Orbitals

GSOs, $|\psi\rangle$, are not eigenstates of the spin operator, $\mathcal{S}_0$, which is defined in second quantization as

$$\mathcal{S}_0 = \sum_{1\leq j\leq s} a^\dagger_{j\alpha} a_{j\alpha} - \sum_{1\leq j\leq s} a^\dagger_{j\beta} a_{j\beta}, \tag{A1}$$

where the fermion second quantized operators $\left\{ a^\dagger_{j\sigma} \middle| 1 \leq j \leq s; \sigma = \alpha, \beta \right\}$ are defined by the action on the vacuum state $|\phi\rangle$ by

$$a^\dagger_{j\sigma} |\phi\rangle = |\varphi_j \sigma\rangle ,$$

where $|\varphi_j\sigma\rangle$ is a product of position and spin states and the anti commutation relationships

$$\left[a_{j\sigma}^{\dagger}, a_{k\nu}\right]_{+} = a_{j\sigma}^{\dagger} a_{k\nu} + a_{k\nu} a_{j\sigma}^{\dagger} = \delta_{jk}\delta_{\sigma\nu}. \tag{A2}$$

GSOs contain non zero components of both alpha and beta spin $\{|\varphi_j\alpha\rangle, |\varphi_j\beta\rangle|\, 1 \leq j \leq s\}$ i.e.

$$|\psi\rangle = \sum_{1\leq j\leq s} \{a_j |\varphi_j\alpha\rangle + b_j |\varphi_j\beta\rangle\}, \tag{A3}$$

which individually are eigenstates of $\mathcal{S}_0$ that satisfy

$$\mathcal{S}_0 |\varphi_j\alpha\rangle = \frac{1}{2}|\varphi_j\alpha\rangle, \quad \mathcal{S}_0 |\varphi_j\beta\rangle = -\frac{1}{2}|\varphi_j\beta\rangle. \tag{A4}$$

2. Pure Spin States

Pure spin states are eigenstates of the spin squared operator, $\mathcal{S}^2$, which is given by

$$\mathcal{S}^2 = \mathcal{S}_-\mathcal{S}_+ + \mathcal{S}_0^2 + \mathcal{S}_0, \tag{A5}$$

where

$$\mathcal{S}_+ = \sum_{1\leq j\leq s} a_{j\alpha}^{\dagger} a_{j\beta}, \tag{A6}$$

$$\mathcal{S}_- = \sum_{1\leq j\leq s} a_{j\beta}^{\dagger} a_{j\alpha} = S_+^{\dagger}. \tag{A7}$$

The action of $\mathcal{S}^2$ on the geminals $|g_\pm\rangle$ is given by

$$\begin{aligned}
\mathcal{S}^2 |g_\pm\rangle &= \mathcal{S}^2 \left(\sqrt{2}\right)^{-1} \{|\varphi_1\alpha\varphi_3\beta\rangle \pm |\varphi_2\alpha\varphi_4\beta\rangle\} = \mathcal{S}_-\mathcal{S}_+ \left(\sqrt{2}\right)^{-1} \{|\varphi_1\alpha\varphi_3\beta\rangle \pm |\varphi_2\alpha\varphi_4\beta\rangle\} \\
&= \mathcal{S}_- \left(\sqrt{2}\right)^{-1} \{|\varphi_1\alpha\varphi_3\alpha\rangle \pm |\varphi_2\alpha\varphi_4\alpha\rangle\} \\
&= \left(\sqrt{2}\right)^{-1} \{|\varphi_1\beta\varphi_3\alpha\rangle + |\varphi_1\alpha\varphi_3\beta\rangle \pm |\varphi_2\beta\varphi_4\alpha\rangle \pm |\varphi_2\alpha\varphi_4\beta\rangle\},
\end{aligned} \tag{A8}$$

which displays that $|g_\pm\rangle$ are not in general eigenstates of $\mathcal{S}^2$. However if one sets $\varphi_3 = \varphi_1$ and $\varphi_4 = \varphi_2$ to form restricted geminals one obtains

$$\mathcal{S}^2 |g_{\mathrm{res},\pm}\rangle = 0, \tag{A9}$$

showing that $|g_\pm\rangle$ are low spin singlet eigenstates of $\mathcal{S}^2$ that factor into a product of space and spin parts i.e.

$$|g_{\mathrm{res},\pm}\rangle = \frac{1}{\sqrt{2}} \{\{\varphi_1(1) \otimes \varphi_1(2) \pm \varphi_2(1) \otimes \varphi_2(2)\} \{\alpha(1) \otimes \beta(2) - \beta(1) \otimes \alpha(2)\}\}. \tag{A10}$$

APPENDIX B: REPRESENTATIVE GSO PAIRS

If the canonical coefficients are nondegenerate, i.e. unequal, the elements of the equivalence classes $[\boldsymbol{\psi}]$ are related to each other by transformations, u with matrix representations $\mathbb{U}$, that belong to the special unitary group $SU(2)$ that transform $\mathbb{C}^2$ to itself, i.e.

$$\{\psi_a, \psi_b\} \sim \{\psi_c, \psi_d\} \Leftrightarrow (\psi_a, \psi_b) = u(\psi_c, \psi_d) = (u\psi_c, u\psi_d) = (\psi_c, \psi_d)\,\mathbb{U}. \tag{B1}$$

This reflects the invariance of geminals $|g\rangle$ to transformations $\tilde{u}$ of one-electron space

$$\tilde{u}\left(|\psi_1\rangle, |\psi_{1+s}\rangle, \cdots, |\psi_s\rangle, |\psi_{s+s}\rangle\right) = \left(|\psi_1\rangle, |\psi_{1+s}\rangle, \cdots, |\psi_s\rangle, |\psi_{s+s}\rangle\right)\left(\begin{pmatrix} \mathbb{U}_1 & \mathbb{O} & \cdots & \mathbb{O} \\ \mathbb{O} & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \mathbb{O} \\ \mathbb{O} & \cdots & \mathbb{O} & \mathbb{U}_s \end{pmatrix}\right), \tag{B2}$$

where $\{\mathbb{U}_j \in 2\times 2 \text{ matrix representation of } SU(2)\,; 1 \le j \le s\}$. The invariance is displayed by

$$\begin{aligned}\Lambda(\tilde{u})\,|g(\boldsymbol{c}, \boldsymbol{\psi}, \boldsymbol{\theta})\rangle &= \sum_{1\le j\le s} c_j e^{i\theta_j} u_j \,|\psi_j\psi_{j+s}\rangle = \sum_{1\le j\le s} c_j e^{i\theta_j}\,|u_j\psi_j u_j\psi_{j+s}\rangle \\ &= \sum_{1\le j\le s} c_j e^{i\theta_j}\left|\psi'_j\psi'_{j+s}\right\rangle = |g(\boldsymbol{c}, \boldsymbol{\psi}, \boldsymbol{\theta})\rangle\,,\end{aligned} \tag{B3}$$

as $\{\psi_j, \psi_{j+s}\} \sim \left\{\psi'_j, \psi'_{j+s}\right\}$ for $1 \le j \le s$, where $\Lambda(\tilde{u})$ is the induced representation of $\tilde{u}$ on two-electron space. In the nondegenerate case the set of all such transformations forms the maximal invariance group of $|g\rangle$.

If degeneracies exist the maximal invariance group and thus the equivalence classes $[\boldsymbol{\psi}]$ are larger, as can be seen from the following discussion. One can expand a geminal in a form that explicitly displays degeneracies as

$$|g(\boldsymbol{c}, \boldsymbol{\psi}, \boldsymbol{\theta})\rangle = \sum_{1\le j\le d} c_j \sum_{1\le k\le d_k} e^{i\theta_{jk}}\left|\psi_{s_{j-1}+k}\psi_{s_{j-1}+k+s}\right\rangle, \tag{B4}$$

where $\{d_1, \cdots, d_d\}$ are the degree of degeneracies of the canonical coefficients $\{c_1, \cdots, c_d\}$, $s_j = \sum_{1\le l\le j} d_l$, and $s_0 = 0$. The simplest example of a degenerate geminal can be realized by setting $s = 3, d = 2, \{d_1, d_2\} = \{2, 1\}$, i.e.

$$|g(\boldsymbol{c}, \boldsymbol{\psi}, \boldsymbol{\theta})\rangle = c_1\left\{e^{i\theta_1}|\psi_1\psi_4\rangle + e^{i\theta_2}|\psi_2\psi_5\rangle\right\} + c_2 e^{i\theta_3}|\psi_3\psi_6\rangle\,. \tag{B5}$$

The equivalence class of pairs for the geminal in eq. $(B5)$ can be divided into two sets

$$[\boldsymbol{\psi}] = [\{\psi_1, \psi_4\}, \{\psi_2, \psi_5\}, \{\psi_3, \psi_6\}] = [\{\{\psi_1, \psi_4\}, \{\psi_2, \psi_5\}\}, \{\psi_3, \psi_6\}]. \tag{B6}$$

The portion of $[\boldsymbol{\psi}]$ associated with $\{\psi_3, \psi_6\}$ can be generated according to eq.$(B1)$, however the portion associated with the degenerate pairs $\{\psi_2, \psi_5\}$ and $\{\psi_3, \psi_6\}$ is more difficult to describe and one must proceed in the fashion outlined in [2, 10, 11]. To this end one defines a set of operators $\{\boldsymbol{u}, \boldsymbol{v}\}$ in terms of second quantized operators $\left\{ a_j^\dagger \middle| j = 1, 2, 4, 5 \right\}$ corresponding to the orthonormal GSOs $\{\psi_j | j = 1, 2, 4, 5\}$ associated with the degenerate canonical coefficient c_1 as

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix} = \begin{pmatrix} e^{-i\theta_1} & 0 & 0 & 0 & 0 & 0 & 0 & e^{-i\theta_2} \\ 0 & e^{-i\theta_1} & 0 & 0 & 0 & 0 & -e^{-i\theta_2} & 0 \\ 0 & 0 & e^{-i\theta_1} & 0 & 0 & -e^{-i\theta_2} & 0 & 0 \\ 0 & 0 & 0 & e^{-i\theta_1} & e^{-i\theta_2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -e^{i\theta_2} & e^{i\theta_1} & 0 & 0 & 0 \\ 0 & 0 & e^{i\theta_2} & 0 & 0 & e^{i\theta_1} & 0 & 0 \\ 0 & e^{i\theta_2} & 0 & 0 & 0 & 0 & e^{i\theta_1} & 0 \\ -e^{i\theta_2} & 0 & 0 & 0 & 0 & 0 & 0 & e^{i\theta_1} \end{pmatrix} \begin{pmatrix} a_1^\dagger a_2 \\ a_1^\dagger a_5 \\ a_4^\dagger a_2 \\ a_4^\dagger a_5 \\ a_2^\dagger a_1 \\ a_5^\dagger a_1 \\ a_2^\dagger a_4 \\ a_5^\dagger a_4 \end{pmatrix}, \tag{B7}$$

where the operators $\boldsymbol{u} = \{u_1, u_2, u_3, u_4\}$, $\boldsymbol{v} = \{v_1, v_2, v_3, v_4\}$ have the properties

$$v\,|g\rangle = 0, \quad u\,|g\rangle \neq 0; \quad u \in \boldsymbol{u}, v \in \boldsymbol{v}. \tag{B8}$$

Using the set $\boldsymbol{v}$ one can form a basis, $\boldsymbol{v}'$, of anti self adjoint operators

$$\begin{aligned} v_1' &= i\left(v_1 + v_1^\dagger\right) \\ v_2' &= v_1 - v_1^\dagger \\ v_3' &= i\left(v_2 + v_2^\dagger\right) \\ v_4' &= v_2 - v_2^\dagger \end{aligned} \tag{B9}$$

that together with the operators

$$\begin{aligned} &a_1^\dagger a_4 - a_4^\dagger a_1, i\left(a_1^\dagger a_4 + a_4^\dagger a_1\right), i\left(a_1^\dagger a_1 - a_4^\dagger a_4\right), \\ &a_2^\dagger a_5 - a_5^\dagger a_2, i\left(a_2^\dagger a_5 + a_5^\dagger a_2\right), i\left(a_2^\dagger a_2 - a_5^\dagger a_5\right) \end{aligned} \tag{B10}$$

form a representation of the unitary symplectic Lie algebra $usp(4)$ [12], which produce a representation of the unitary symplectic group $USp(4)$ [12] over the subspace generated by $\{\psi_1, \psi_2, \psi_4, \psi_5\}$, that leaves $|g\rangle$ invariant. Thus any GSO produced by the action of this representation of $USp(4)$ produces the same geminal, hence all the GSOs $\{USp(4)\,|\boldsymbol{\psi}'\rangle \equiv USp(4)\,(|\psi_1\rangle, |\psi_2\rangle, |\psi_4\rangle, |\psi_5\rangle)\}$ belong to the same equivalence class $[\boldsymbol{\psi}]$, which generalizes the equivalence relationship given in eq. (10). The set $\boldsymbol{u}'$, formed in an analogous manner to the set $\boldsymbol{v}'$, is contained in a representation of the Lie algebra $u(4)$ that produces a representation of the group $U(4)$ that changes $|g\rangle$

APPENDIX C: SORDO

Consider the SORDO of the geminal $|g_-\rangle$

$$D^2(g_-) = \frac{1}{2}\{|1\alpha 3\beta\rangle\langle 1\alpha 3\beta| - |1\alpha 3\beta\rangle\langle 2\alpha 4\beta| - |2\alpha 4\beta\rangle\langle 1\alpha 3\beta| + |2\alpha 4\beta\rangle\langle 2\alpha 4\beta|\}, \tag{C1}$$

where

$$m = \frac{N}{2} = \frac{2}{2} = 1, \tag{C2}$$

$$s = 2, \tag{C3}$$

$$n_1 = n_2 = \frac{1}{2}, \tag{C4}$$

$$c_1 = c_2 = \frac{1}{\sqrt{2}}, \tag{C5}$$

$$\theta_1 = 0, \quad \theta_2 = \pi. \tag{C6}$$

Substituting for m, s and r in the general form for the SORDO results in

$$D^2(g_-) = (S_1)^{-1}\left\{\sum_{1\le j\le 2} n_j S_0(\hat{\jmath})\,|\psi_j\psi_{j+s}\rangle\langle\psi_j\psi_{j+s}| \right.$$
$$\left. + \sum_{1\le j\ne k\le 2} c_j c_k e^{i(\theta_j-\theta_k)} S_0\left(\hat{\jmath}\hat{k}\right)|\psi_j\psi_{j+s}\rangle\langle\psi_k\psi_{k+s}| + \sum_{\substack{1\le j<k\le 4\\ k\ne j+2}} n_j n_k S_{-1}\left(\hat{\jmath}\hat{k}\right)|\psi_j\psi_k\rangle\langle\psi_j\psi_k| \right\}. \tag{C7}$$

Noting that the symmetric functions are given by

$$S_1 = n_1 + n_2 = 1 \tag{C8}$$

$$S_0\left(\hat{1}\hat{2}\right) = 1 \tag{C9}$$

$$S_0\left(\hat{1}\right) = S_0\left(\hat{2}\right) = 1 \tag{C10}$$

$$S_{-1}\left(\hat{1}\hat{2}\right) = 0. \tag{C11}$$

and substituting for $\boldsymbol{c}, \boldsymbol{n}$ and $\boldsymbol{\theta}$ one obtains

$$D^2\left(g_-\right) = \frac{1}{2}\left\{\left|\psi_1\psi_3\right\rangle\left\langle\psi_1\psi_3\right| + \left|\psi_2\psi_4\right\rangle\left\langle\psi_2\psi_4\right|\right\} + \frac{1}{2}\left\{\left|\psi_1\psi_3\right\rangle\left\langle\psi_2\psi_4\right| e^{-i\pi} + \left|\psi_2\psi_4\right\rangle\left\langle\psi_1\psi_3\right| e^{i\pi}\right\} \tag{C12}$$

$$= \frac{1}{2}\left\{\left|\psi_1\psi_3\right\rangle\left\langle\psi_1\psi_3\right| - \left|\psi_1\psi_3\right\rangle\left\langle\psi_2\psi_4\right| - \left|\psi_2\psi_4\right\rangle\left\langle\psi_1\psi_3\right| + \left|\psi_2\psi_4\right\rangle\left\langle\psi_2\psi_4\right|\right\} \tag{C13}$$

and finally substituting for $\boldsymbol{\psi}$ one gets eq. $(C1)$.

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